

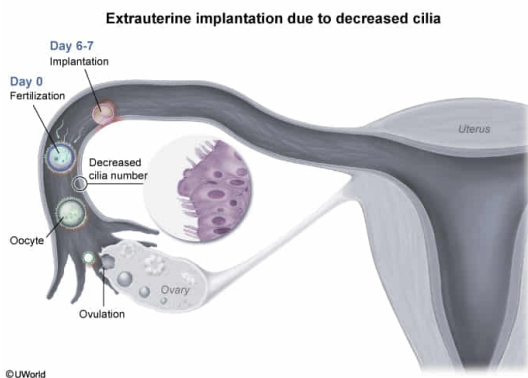
Which of the following would most likely cause an embryo to implant in a location other than the uterine lining?

- ✓ ☒ A. Reduced number of fallopian cilia [63%]
☐ B. Surge in luteinizing hormone prior to ovulation [9%]
☐ C. Incomplete gastrulation [10%]
☐ D. Rupture of follicle from the ovary [15%]

Correct

64% answered correctly

Explanation:



Oogenesis is the process by which females produce sex cells (gametes) called eggs. All of a woman's **developing eggs (oocytes)** are produced during fetal development. At birth, their maturation is arrested in **prophase I**; these arrested eggs are known as **primary oocytes**. At puberty, the **menstrual cycle** begins, and each month a single primary oocyte develops into a **secondary oocyte** by continuing **meiosis** up to **metaphase II**.

During the follicular phase of the cycle, the follicle (oocyte plus supporting granulosa cells) matures until it bulges at the outer wall of the ovary. Maturation and growth of the follicle is driven by the release of follicle-stimulating hormone (FSH) from the **anterior pituitary**. A subsequent release of luteinizing hormone (LH) stimulates follicle rupture and the release of the secondary oocyte near the opening of the fallopian tube (**ovulation**).

Fertilization (joining of male and female gametes into a zygote) becomes possible after the released secondary oocyte has been drawn into the fallopian tube. It is normal for fertilization to occur in the fallopian tube prior to implantation within the uterus. **Typically**, small hair-like structures within the fallopian tube called **fallopian cilia** help **propel** the fertilized oocyte toward the uterus for implantation. However, with a **reduced number of fallopian cilia**, improper implantation of the fertilized egg can occur **outside the uterus**.

(Choices B and D) A surge in LH and follicular rupture from the ovary are necessary steps for ovulation to occur. LH provides the necessary stimulus for follicular rupture and the release of a secondary oocyte into the fallopian tube. Because these steps must happen *before* fertilization can occur, they would *not* be the cause of embryonic attachment outside the uterus.

(Choice C) **Gastrulation** is a post-implantation process by which a developing embryo

progresses from the blastula stage to the tri-layered gastrula stage. Because this process occurs post-implantation, abnormal gastrulation would *not* cause implantation outside the uterus.

Educational objective:

After fertilization, fallopian cilia help propel the fertilized oocyte toward the uterus for implantation. An inadequate number of cilia in the fallopian tube can cause implantation of the fertilized egg outside the uterus.

Time spent:0 sec

Subject:
Biology

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System:
Reproduction